Sanaul Haque

Illari_Anne_Cerna_Quiliche_Proposal

Course 66

Document Details

Submission ID
trn:oid:::3117:636202558

Submission Date
Oct 8, 2026, 11:45 PM GMT+6

Download Date
Oct 8, 2026, 11:48 PM GMT+6

File Name
Illari_Anne_Cerna_Quiliche_Proposal.pdf

File Size
288.3 KB

8 Pages

2,122 Words

12,409 Characters

4% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **6** Not Cited or Quoted 3%
  Matches with neither in-text citation nor quotation marks
- **1** Missing Quotations 0%
  Matches that are still very similar to source material
- **2** Missing Citation 1%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

4% Internet sources
2% Publications
1% Submitted works (Student Papers)

Match Groups

- **6** Not Cited or Quoted 3%
  Matches with neither in-text citation nor quotation marks
- **1** Missing Quotations 0%
  Matches that are still very similar to source material
- **2** Missing Citation 1%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

4% Internet sources
2% Publications
1% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	unhabitat.org 1%
2	Internet	www.forosperu.net <1%
3	Student papers	National University College - Online on 2026-09-12 <1%
4	Internet	www.coursehero.com <1%
5	Internet	larepublica.pe <1%
6	Internet	pesquisa.bvsalud.org <1%
7	Internet	www.enfoquederecho.com <1%

EXPLORATORY RESEARCH PROPOSAL

ON

EDENA

BIODEGRADABLE FUNERARY CAPSULES VERSUS CEMETERY LAND USE AND THE GREEN PUBLIC SPACE DEFICIT IN CHIMBOTE AND NUEVO CHIMBOTE

An exploration of its environmental, economic, and social viability

Chimbote, Áncash — Peru

PROPONENT:

ILLARI

Date of preparation: September 2026

File name: EDENA_Research_Proposal_IEEE_EN.docx

Word count: ≈ 2,049

Character count: ≈ 15,215

RESEARCH PROPOSAL
ON

EDENA

A SUSTAINABLE BIODEGRADABLE FUNERAL CAPSULE SYSTEM AS AN ALTERNATIVE TO THE LACK OF URBAN GREEN INFRASTRUCTURE

Chimbote, Áncash — Perú



PROPONENT:
ILLARI

Date of preparation: September 2026

1. The Need for Green Infrastructure

Urban green areas regulate temperature, improve air quality, capture carbon, and support physical and mental health. The World Health Organization (WHO) recommends 9–15 m² of green space per inhabitant, a figure consistently cited in Peruvian urban studies such as Bustamante Sánchez [1] on Chiclayo.

For Chimbote and Nuevo Chimbote, an official local measurement also exists. The Voluntary Local Review on SDG 11 compliance, prepared by Foro Ciudades para la Vida with the Provincial Municipality of Santa, Instituto Natura, and UN-Habitat, found that both districts together had 1,011,778 m² of open public space (plazas and local, sectoral, and zonal parks) for a population of 365,534 — 2.77 m² per person, less than a third of the WHO minimum [2]. The report also breaks the indicator down by age group: only 0.69 m² per person for ages 1–14, and 0.26 m² for those over 65.

The city's own 2020–2030 Urban Development Plan (PDU), cited in that report, projects a deficit of approximately 50,000 m² in plaza infrastructure and 680,000 m² in sectoral park infrastructure by 2030, with no deficit identified in local or zonal parks [2]. Chimbote's green-space deficit is therefore not a generalization borrowed from other cities: it is officially acknowledged by the Provincial Municipality of Santa within its own urban planning instrument.

2. Problem Statement

A possible tension exists between two processes coexisting in the city: Chimbote's urban planning instruments acknowledge, with their own figures, the need to expand green public space, while current funerary practice continues to depend on single-use wooden coffins and the progressive expansion of permanently occupied cemetery land. This research does not assume the two processes compete directly for the same land; it proposes to examine to what extent that relationship actually exists.

2.1 Municipalities and Welfare Boards: What Problem Do They Face?

Law No. 26298, the Cemeteries and Funeral Services Law, requires every public cemetery to allocate at least 15% of its area to common-ground burials and an additional 5% to free burials and mass graves [3]. Any future expansion of a public cemetery in Chimbote therefore carries a legally mandated land commitment.

Chimbote's main cemetery — Divino Maestro, with 97 pavilions — is managed by the Sociedad de Beneficencia Pública de Chimbote, which reported over 20,000 visitors in a single day during All Saints' Day [4], [5]. That figure shows the scale of use but is not, by itself, evidence of saturation: no public information was found on remaining capacity, unclaimed bodies, annual maintenance costs, or expansion plans.

This is accordingly one of the central questions this research seeks to answer through direct consultation with both institutions, rather than something assumed from information available so far. If expansion is confirmed necessary, it would, in principle, compete for the same urban land the

PDU 2020-2030 already identifies as deficient for plazas and sectoral parks — but that relationship must be verified, not assumed.

2.2 The Funeral Industry and Wood Consumption: A Hypothesis to Verify

The Italian project Capsula Mundi — which proposes replacing coffins with a biodegradable capsule — argues that manufacturing a coffin today requires felling a mature tree for an object with a useful life of only a few days, when that tree would have taken 10 to 40 years to grow [6]. This is the central argument of that initiative, not a verified fact about Chimbote.

No specific evidence was found on the type of wood used in coffins sold in Chimbote, the average volume per unit, its origin, or annual sales. The link between the local funeral industry and tree felling remains, for now, a working hypothesis drawn from international experience rather than a proven local fact. Verifying it through direct consultation with local carpentry shops, funeral homes, and suppliers is part of this study's planned work; its current absence is explicitly acknowledged as a limitation (see Section 8).

2.3 Who Needs EDENA?

Two user groups follow from this diagnosis and are worth distinguishing. The first is institutional: the Provincial Municipality of Santa and the Sociedad de Beneficencia Pública de Chimbote manage cemetery land that the city also needs to close the green-space deficit its own PDU 2020-2030 acknowledges. A model that permanently reduces the land committed per burial could, in principle, ease that pressure.

The second group is Chimbote and Nuevo Chimbote families, who must cover a funeral service — between S/ 1,400 and S/ 3,700 by general Peruvian figures — plus a burial plot that can exceed S/ 20,000, against an EsSalud burial subsidy that does not exceed S/ 2,070 [7], [8]. This group faces, above all, a problem of economic access rather than land management.

These are not necessarily the same problem, though they are linked through the Municipality and the Beneficencia, which manage the land and set much of the funeral service conditions. EDENA does not assume a single proposal automatically solves both: this research proposes to examine separately how the model would benefit each group, and whether the two benefits could reinforce one another.

3. Controversy

The question guiding this research does not start from a predetermined answer: to what extent does the traditional funerary model — based on single-use wooden coffins and permanent cemetery-land expansion — actually compete with the need for green public space and with families' economic capacity in Chimbote and Nuevo Chimbote, and to what extent could an alternative model such as EDENA ease that competition without creating new environmental, health, or cultural problems?

This question allows for more than one answer. The research could find that land competition is smaller than assumed, that the local funeral industry's wood consumption is marginal compared to other causes of green-area loss, or that regulatory, health, or cultural barriers make a model like EDENA unfeasible in the short term — or it could confirm the initial hypothesis,

wholly or in part. In any case, the purpose is to generate evidence to decide on, not to confirm an already-assumed conclusion. Precedents such as Capsula Mundi [6] and recycled cardboard coffins — around 35 euros in the Spanish market versus 300–2,000 euros for a conventional coffin [9] — serve here as motivation, not as proof that the alternative is already valid for Chimbote.

4. Research Questions

- What is the actual relationship between cemetery-designated land in Chimbote and the green public space deficit acknowledged in the PDU 2020-2030?
- What evidence exists on the local funeral industry's wood consumption, and to what extent does it compete with the city's urban forest resources?
- What concrete problems do the Sociedad de Beneficencia Pública de Chimbote and the Provincial Municipality of Santa face in managing the Divino Maestro cemetery (available space, unclaimed bodies, maintenance costs, expansion plans)?
- To what extent would a system like EDENA be more economically accessible than current funeral services, and which user group — families or administering institutions — would benefit more directly?
- What is the level of acceptability among the population, funeral homes, and cemetery-administering entities toward an ecological funerary model such as EDENA?

5. Comparative Table: Traditional Model versus EDENA

Criterion	Traditional Funerary Model	Proposed Model (EDENA)
Main material	Wooden coffin (hypothesis to verify locally) and concrete/marble niche	Biodegradable material of plant origin
Cemetery land use	Permanently occupied pavilions/niches, subject to the legal minimum of 15% + 5% (Law No. 26298)	Land that could, in theory, evolve into plant cover (to be validated)
Approx. cost (Peru reference, not Chimbote-specific)	S/ 1,400–3,700 for the service, plus S/ 20,000+ for the plot	Estimated lower cost by removing the coffin and permanent niche (to be defined at prototype stage)
Environmental contribution	To be confirmed: depends on wood volume/origin in Chimbote, data not yet available	Potentially positive, subject to validating behavior in Chimbote's soil and climate

Note: cost and legal-framework rows rely on documented, cited sources; material and environmental-contribution rows describe hypotheses pending local verification, not established facts for Chimbote.

6. Proposed Methodology

The approach is mixed, combining local fieldwork, documentary analysis, and experimental development, in the following phases:

- Institutional verification: direct interviews with the Provincial Municipality of Santa and the Sociedad de Beneficencia Pública de Chimbote on cemetery capacity, unclaimed bodies, management costs, and expansion plans.
- Wood-consumption study: consultation with local carpentry shops and funeral homes on the type, volume, and origin of wood used in coffins sold in Chimbote.
- Comparative cost study between the traditional model and EDENA, using data gathered in the previous phases.
- Prototype development: design and 3D-printed fabrication of the biodegradable EDENA capsule, including validation of its behavior (degradation rate, absence of leaks or odors) under Chimbote's desert-coastal soil and climate.
- Acceptability assessment, applied separately to families and institutional representatives (funeral homes, Beneficencia, Municipality).
- Analysis and recommendations, synthesizing results from the previous phases.

7. Expected Outcome

This research is expected to determine, with evidence gathered directly in Chimbote, whether real competition exists between cemetery land use and the city's need for green public space, how significant the local funeral industry's wood consumption is, and whether a model like EDENA represents a viable alternative — environmentally, economically, and culturally — for one or both user groups identified in Section 2.3.

8. Potential Limitations

- Current Peruvian cemetery regulations were designed for traditional burial models, so EDENA could face legal gaps or require regulatory adaptation.
- Cultural or religious resistance to non-conventional forms of farewell may exist and will need direct exploration during the acceptability assessment.
- No local evidence yet exists on the type and volume of wood actually used in Chimbote's coffins; generating that data is part of this research's challenge, not a starting assumption.
- No evidence exists on how a biodegradable capsule behaves under Chimbote's specific soil, humidity, and climate; validating that behavior is part of the project's experimental component.

9. Conclusion

EDENA starts from a question, not an already-decided answer. This research proposes to examine whether the traditional funerary model actually competes with the need for green public space that the Provincial Municipality of Santa itself acknowledges in the PDU 2020-2030, and whether a biodegradable capsule system could offer a viable alternative to that tension and to the economic cost families face when saying goodbye to a loved one. It aims to contribute local evidence — environmental, economic, and on social acceptability — that does not yet exist in

systematized form, enabling the Municipality, the Beneficencia, and the public to make better-informed decisions about the future of funeral services.

10. Project Practicalities

Development is estimated at approximately eight months: the first two months will cover institutional and regulatory verification and the wood-consumption study; the next four months, prototype development, validation, and acceptability surveys; and the final two months, results analysis and the final report.

REFERENCES

[1] S. V. Bustamante Sánchez, "Infraestructura verde productiva aplicada en el Sector XI de Chiclayo para mitigar el déficit de espacios verdes," Tesis, Universidad Católica Santo Toribio de Mogrovejo, 2025. [Online]. Available: https://repositorio.usat.edu.pe/bitstreams/a5cbfc28-f50d-47e4-9261-889380fb4dd7/download. Accessed: Sep. 2026.

[2] M. E. Foronda Farro and L. Miranda Sara, "Informe Local Voluntario sobre el cumplimiento del ODS 11 en las ciudades de Chimbote y Nuevo Chimbote," Foro Ciudades para la Vida, Municipalidad Provincial del Santa e Instituto Natura, en colaboración con ONU-Hábitat, 2020. [Online]. Available: https://unhabitat.org/sites/default/files/2021/06/chimbote_2020_es.pdf. Accessed: Sep. 2026.

[3] Congreso de la República del Perú, Ley N.° 26298, Ley de Cementerios y Servicios Funerarios, y su Reglamento, aprobado por Decreto Supremo N.° 03-94-SA, 1994.

[4] Diario Andina, "Con ambiente festivo chimbotanos recuerdan a sus familiares en cementerio," Andina, Nov. 1, 2019. [Online]. Available: https://andina.pe/agencia/noticia-con-ambiente-festivo-chimbotanos-recuerdan-a-sus-familiares-cementerio-771697.aspx. Accessed: Sep. 2026.

[5] Diario Andina, "Día de Todos los Santos: más de 20,000 personas visitaron cementerio de Chimbote," Andina, Nov. 1, 2019. [Online]. Available: https://andina.pe/agencia/noticia-dia-todos-los-santos-mas-20000-personas-visitaron-cementerio-chimbote-961311.aspx. Accessed: Sep. 2026.

[6] P. Carr-Gomm, "Capsula Mundi," n.d. [Online]. Available: https://www.philipcarr-gomm.com/capsula-mundi/. Accessed: Sep. 2026.

[7] La República, "¿Cuánto cuesta enterrar a un muerto en Perú? Funerarios y sepelio en camposantos," n.d. [Online]. Available: https://larepublica.pe/economia/1348462-cuesta-enterrar-muerto-peru-funerarios-sepelio-campo-santo. Accessed: Sep. 2026.

[8] La República, "EsSalud: este es el monto que entrega el seguro en Perú por gastos funerarios si fallece un familiar asegurado," Feb. 19, 2025. [Online]. Available: https://larepublica.pe/economia/2025/02/19/essalud-este-es-el-monto-que-entrega-el-seguro-en-peru-por-gastos-funerarios-si-fallece-un-familiar-asegurado-atmp-582692. Accessed: Sep. 2026.

[9] Consumer Eroski, "Ataúdes ecológicos," Jun. 16, 2006. [Online]. Available: https://www.consumer.es/?p=27328. Accessed: Sep. 2026.